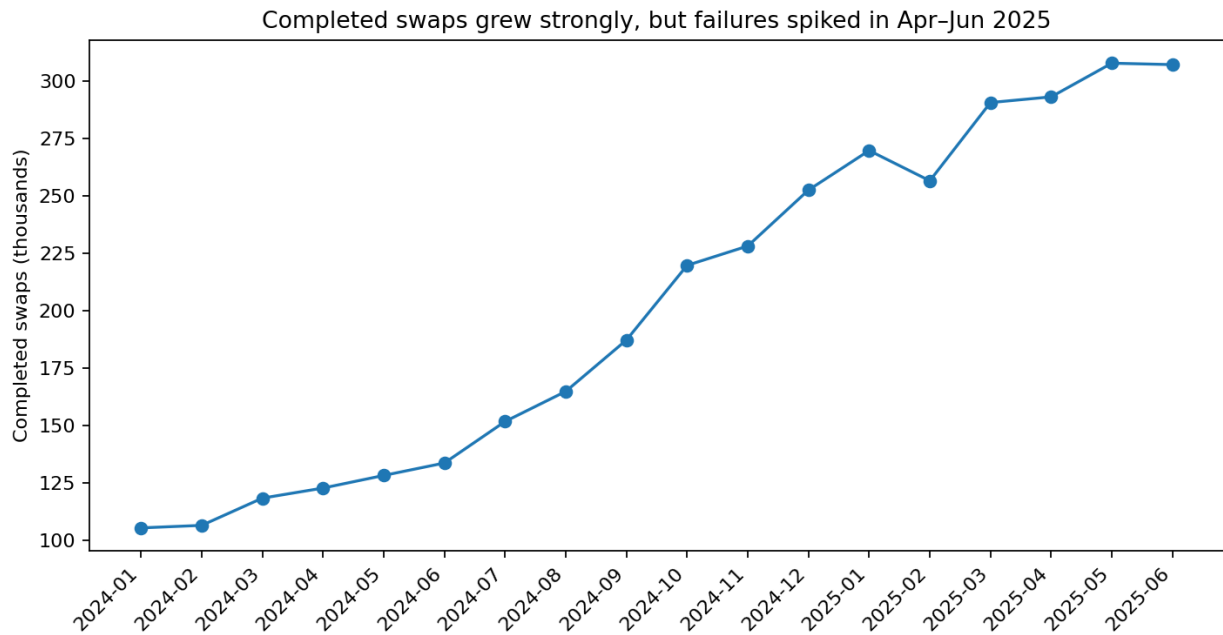


VoltRelay Energy — Final Data Analytics Report

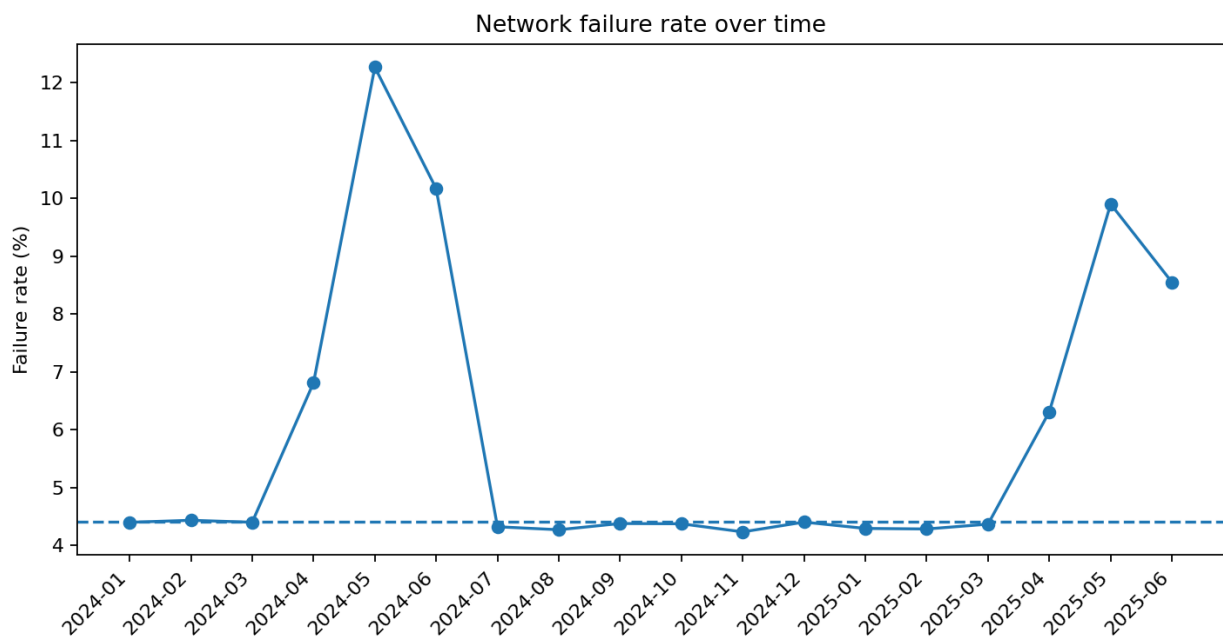
Gradient Learnings Data Analytics Hackathon

January 2024 – June 2025 | 3,877,013 swap attempts | 6 cities | 152 stations

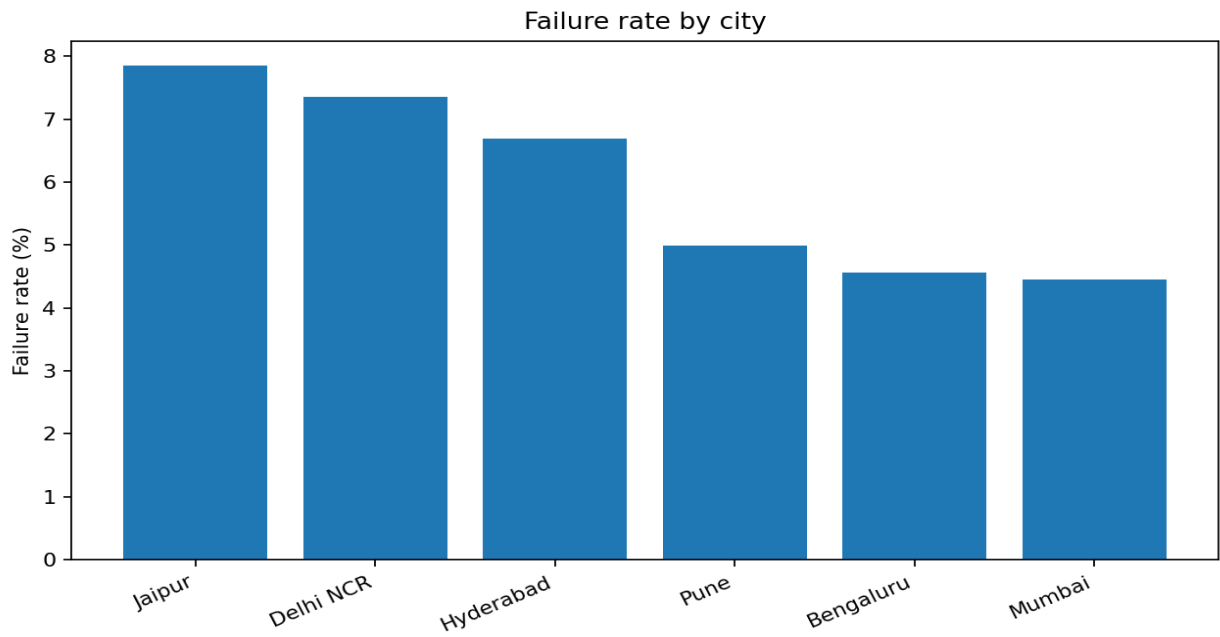
Executive summary. VoltRelay is growing rapidly, but reliability is deteriorating in concentrated station, city and peak-hour pockets. The evidence points primarily to charged-battery availability and queue abandonment, with Gen1 station performance and weak battery cohorts as important secondary signals. Retention is associated with first-month failure exposure, while the strongest commercial issue is a localized discount step-up at FP-03.



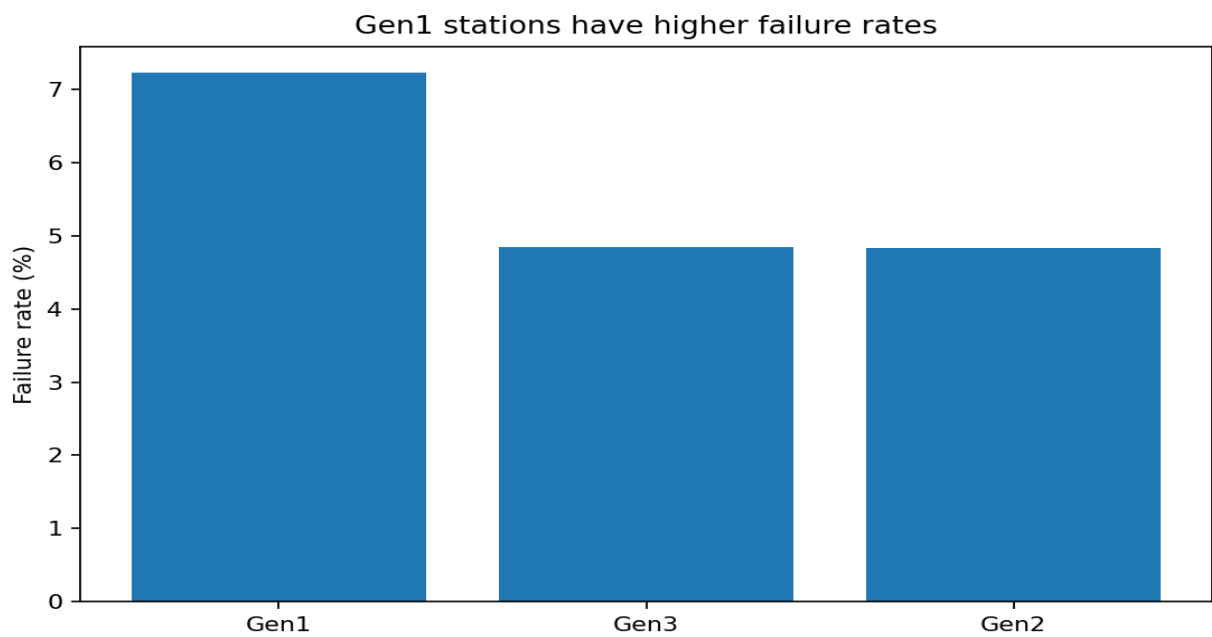
Growth. Completed swaps rose from about 105k in Jan-2024 to 307k in Jun-2025. Overall failure rate is 5.96%, with spikes to 12.27% in May-2024 and 9.91% in May-2025.



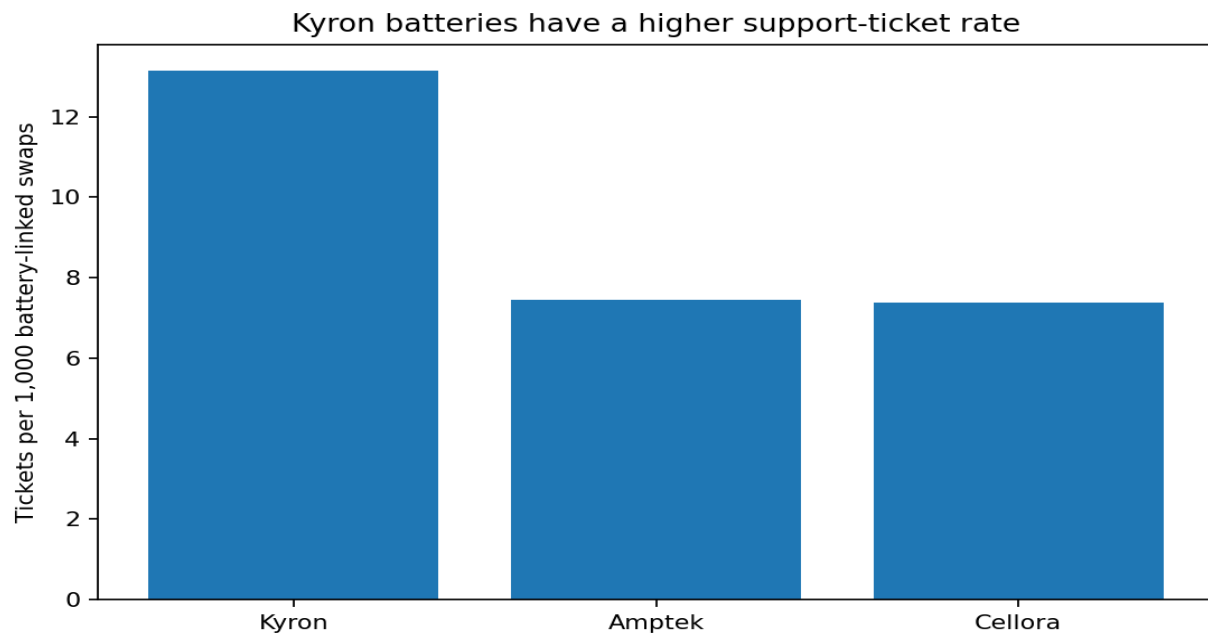
Failure mix. 58.1% of failures are failed-no-charged-battery events and 29.6% are abandoned queues. System errors are only about 5% of failures.



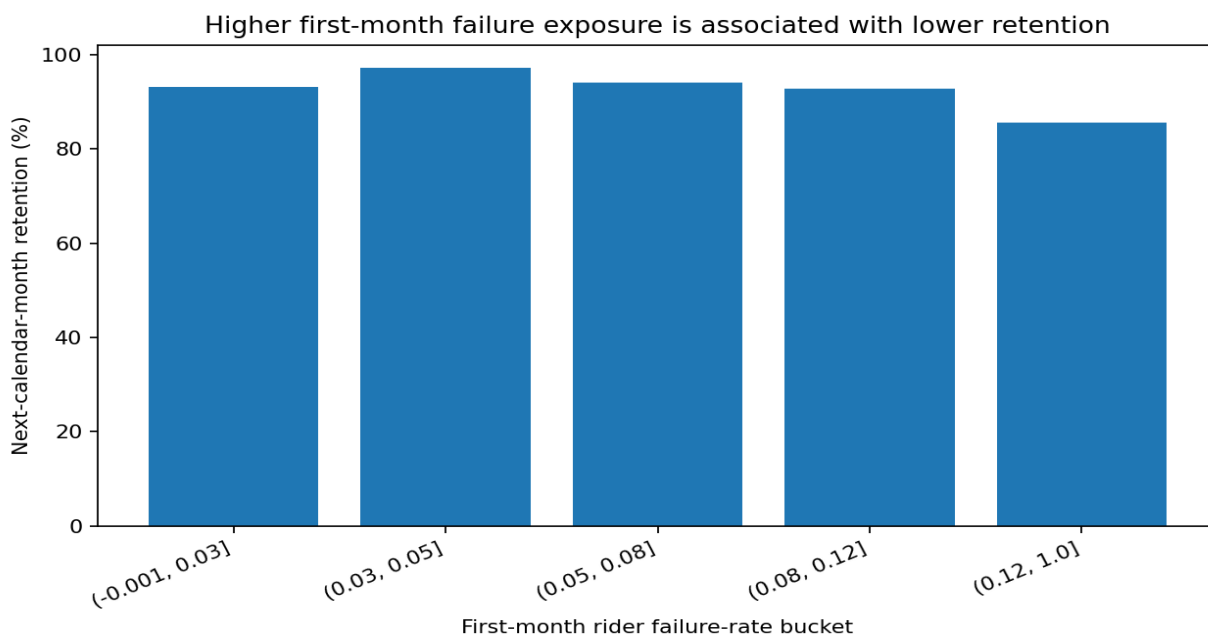
Geography. Jaipur and Delhi NCR have the highest city-level failure rates. The top 20 stations account for about 30% of failures, supporting targeted intervention.



Equipment. Gen1 stations have an observed failure rate around 7.17% versus about 4.87% for Gen2 and 4.88% for Gen3. This is an association, not causal proof.



Battery. Kyron has about 13.1 battery-linked tickets per 1,000 swaps versus about 7.4 for Cellora and Amptek; Kyron lots 2407–2409 are particularly low-SOH cohorts.



Retention. Riders with first-month failure exposure above 12% have about 85.5% next-month retention, versus about 93% at 3% or below. This is descriptive, not causal.

Recommendations. (1) Pre-position batteries and manage 19:00–22:00 peaks at high-risk stations; (2) retrofit high-volume Gen1 sites before blanket expansion; (3) quarantine/rotate weak battery cohorts; (4) launch first-30-day rider recovery; (5) review FP-03 discount economics; (6) operate a weekly station-health dashboard.

Margin note. The raw dataset has no direct contribution-margin field. The analysis therefore uses a transparent operating contribution proxy: realized revenue minus energy cost and allocated station rent/maintenance. Under this proxy, network-wide margin does not show broad erosion; the strongest commercial leakage is localized at FP-03.

Limitations. Synthetic data; observational associations; missing-not-at-random CSAT; contribution margin proxy rather than accounting margin; explicit retention definition.